

# Aether — SLA Compliance Monitoring

Uptime Targets, Error Budgets, and Compliance Reports

Define SLA tiers per workload, track compliance against uptime, latency, error rate, and restart metrics, and receive proactive violation alerts.

Best-Effort • Standard • High-Availability

# SLA Tiers

Aether provides three pre-built SLA tiers with configurable targets for uptime, latency, error rate, and restart count. Custom tiers can be defined per workload.

| Metric                  | Best-Effort       | Standard  | High-Availability |
|-------------------------|-------------------|-----------|-------------------|
| Uptime Target           | 99.0%             | 99.9%     | 99.99%            |
| Max Latency             | 2,000 ms          | 500 ms    | 200 ms            |
| Max Error Rate          | 5.0%              | 1.0%      | 0.1%              |
| Max Restarts/Day        | 10                | 3         | 1                 |
| Monthly Downtime Budget | ~432 min (7.2 hr) | ~43.2 min | ~4.32 min         |



## Best-Effort

For development and internal tools. 99% uptime allows ~7 hours of downtime per month. Generous error rate and restart limits support rapid iteration.



## Standard

For production services with business impact. 99.9% uptime allows ~43 minutes per month. Sub-second latency targets ensure responsive user experiences.



## High-Availability

For mission-critical services. 99.99% uptime allows only ~4 minutes of downtime per month. Strictest latency and error rate requirements.

```
# Set SLA tier for a workload
$ aether sla set my-app --tier standard
```

```
$ aether sla set payment-api --tier high-availability  
$ aether sla set dev-tools --tier best-effort
```

# Compliance Checking

Evaluate workload metrics against SLA targets with a single command. Aether checks uptime, latency, error rate, and restart count, then reports pass/fail with margin percentages.

```
$ aether sla check my-app
```

```
SLA Report:  my-app
```

```
Status:      At Risk
```

| Metric       | Target | Actual | Status | Margin |
|--------------|--------|--------|--------|--------|
| Uptime       | 99.90% | 99.92% | PASS   | 0%     |
| Latency      | 500ms  | 420ms  | PASS   | 16%    |
| Error Rate   | 1.00%  | 0.85%  | PASS   | 15%    |
| Restarts/Day | 3      | 1      | PASS   | 67%    |

```
Error Budget:  21.6 min remaining of 43.2 min (50% consumed)
```

```
Projected exhaustion: 15 days
```

```
Recommendations:
```

```
  Uptime is within 0% of SLA limit - monitor closely
```

## Compliance Statuses



### Compliant

All metrics pass their SLA targets with comfortable margins. No action required. Error budget consumption is on track.



### At Risk

All metrics currently pass, but one or more is within 10-15% of the SLA limit. The uptime margin is below 0.1%. Proactive



### Violated

One or more metrics have breached their SLA target. Immediate remediation required. Error budget may be exhausted.

intervention  
recommended.

## Error Budget Tracking

The error budget is calculated from the SLA uptime target:  $(100\% - \text{target\%}) * \text{minutes\_per\_month}$ . For a 99.9% SLA, the total budget is 43.2 minutes/month. Aether tracks consumed minutes based on observed downtime and projects when the budget will be exhausted at the current burn rate. Teams can use this projection to make informed decisions about feature velocity vs. reliability investment.

# SLA Violation Alerts

When a workload's metrics breach SLA targets, Aether generates actionable recommendations and surfaces violation alerts through the health monitoring system.

## Alert-Triggered Recommendations

| Metric Violated | Recommendations                                                                                          |
|-----------------|----------------------------------------------------------------------------------------------------------|
| Uptime          | Enable health probes and auto-restart for higher uptime. Consider adding replica scaling for redundancy. |
| Latency         | Profile application for performance bottlenecks. Consider adding caching layer or CDN.                   |
| Error Rate      | Analyze error logs to identify root causes. Implement circuit breaker patterns.                          |
| Restarts/Day    | Investigate OOM kills or crash loops. Review resource limits and liveness probe timing.                  |

```
$ aether sla check payment-api

SLA Report: payment-api
Status: VIOLATED

Metric      Target    Actual    Status    Margin
-----
Uptime      99.99%    99.50%    FAIL      -0%
Latency     200ms     180ms     PASS      10%
Error Rate  0.10%     2.30%     FAIL      -2200%
Restarts/Day 1          4         FAIL      -300%

Error Budget: 0.0 min remaining of 4.3 min (100% consumed)
Projected exhaustion: EXHAUSTED

Recommendations:
```

- Enable health probes and auto-restart for higher uptime
- Consider adding replica scaling for redundancy
- Analyze error logs to identify root causes
- Implement circuit breaker patterns
- Investigate OOM kills or crash loops
- Review resource limits and liveness probe timing

# Health & **Webhook** Integration

SLA compliance monitoring integrates with Aether's health monitoring system and supports webhook notifications for SLA breaches.



## Health Probe Integration

SLA uptime is derived from health probe results. When a workload's liveness or readiness probe fails, the downtime is tracked against the SLA error budget in real time.



## Metrics Collection

Latency and error rate metrics flow from Aether's built-in metrics system. Restart counts are tracked directly from runtime events across Podman, Kubernetes, KubeVirt, and Metal3.



## Webhook Notifications

Configure webhook endpoints to receive alerts when SLA status transitions from Compliant to At Risk or Violated. Payloads include the full SLA report with all metrics and recommendations.



## Compliance Reports

Export SLA compliance reports for all workloads. Reports include historical uptime, error budget consumption trends, and projected exhaustion dates for management and compliance review.

## Webhook Configuration

```
# Configure SLA violation webhook
$ aether sla webhook --url https://hooks.slack.com/aether-sla \
  --events violated,at-risk

# Webhook payload example
{
  "workload": "payment-api",
```

```
"status": "VIOLATED",
"checks": [
  { "metric": "Uptime", "target": "99.99%", "actual": "99.50%", "passed":
  { "metric": "Error Rate", "target": "0.10%", "actual": "2.30%", "passed
],
"error_budget": { "remaining_minutes": 0.0, "consumed_pct": 100.0 },
"timestamp": "2026-04-12T14:30:00Z"
}
```

*SLA compliance is not a dashboard you check — it is an alert system that tells you before your customers do. Aether makes SLA monitoring proactive, not reactive.*

# Meet Your **SLAs**

Define targets, track compliance, and receive alerts before budgets are exhausted.

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Pre-Built SLA Tiers

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Compliance Metrics

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Compliance Statuses

## SLA Compliance Built Into the Control Plane

Uptime, latency, error rate, and restart tracking with error budget projections and webhook alerts — across Podman, Kubernetes, KubeVirt, and Metal3.

```
# Complete SLA workflow
$ aether sla set my-app --tier standard
$ aether sla check my-app
$ aether sla webhook --url https://hooks.slack.com/alerts
$ aether sla report --all --format json
```